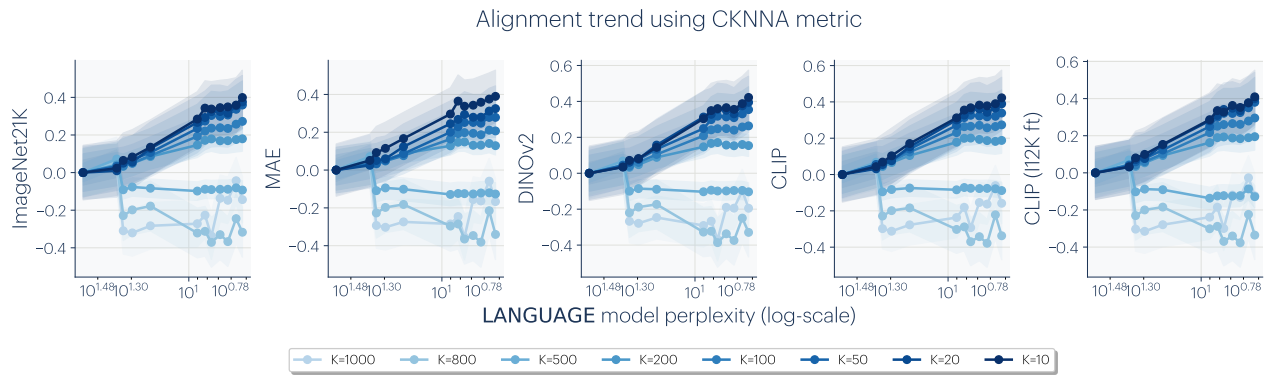

Figure 10. **Cross-modal alignment increases locally:** Alignment trend when varying the top-k nearest neighbors in the CKNNA metrics (Eqn. 18). We center alignment score to the smallest language model and divide the total trend by the standard deviation. When $k = 1024$, we recover the original CKA metric, and when $k < |\mathcal{X}|$ it closely resembles the mutual nearest-neighbor metric m_{NN}. Each line represents the average of all LLM models for a specific k. As we decrease k, the alignment becomes more pronounced.

Where $\alpha(i, j)$ is a scalar weighting that assigns 1 if j is a mutual nearest neighbors to both ϕ_i and ψ_i, and 0 otherwise. We refer to this metric as the Centered Kernel Nearest-Neighbor Alignment (CKNNA) metric. As the number of nearest neighbors $k \to \dim(\mathbf{K})$, we recover the original CKA metric.

$$\mathsf{CKNNA}(\mathbf{K}, \mathbf{L}) = \frac{\mathsf{Align}_{\mathsf{knn}}(\mathbf{K}, \mathbf{L})}{\sqrt{\mathsf{Align}_{\mathsf{knn}}(\mathbf{K}, \mathbf{K}), \mathsf{Align}_{\mathsf{knn}}(\mathbf{L}, \mathbf{L})}} \tag{18}$$

We can further relax the metric to treat the cross-covariance term identically across all nearest-neighbor samples. This is equivalent to the assumption that all nearby samples have the same distance. This simplification leads us back to the mutual nearest neighbor metric:

$$\sum_i \sum_j \alpha(i, j) \cdot 1 = n \cdot k \cdot m_{\mathsf{NN}}(\phi_i, \psi_i) \tag{19}$$

By equating these metrics, we analyze the changes in alignment between language and vision models as we vary the number of neighbors k in Eqn. 18. In Figure 10, we compute the average alignment score across all LLM models. For each k, we center the scores to the smallest vision model and divide by the standard deviation of the scores. We find that high values of k show less conclusive alignment across tasks while decreasing k shows a coherent trend across both models and tasks.

B. Consistency across various metrics

We describe the metrics in Table 11 and their corresponding properties. The *symmetric* property implies that the metric is symmetric with respect to the data points $d(x, y) = d(y, x)$. The *global* property means all samples are used to compute the distance with respect to every sample. The *ordinal* property is when the ordering of the distance is taken into consideration. For example, mutual nearest neighbor is not ordinal since the nearest neighbors $\{a, b, c\}$ and $\{c, a, b\}$ are treated equally. The *batchable* property is a computational property that makes it feasible to compute in a reasonable time frame.

Vision-vision comparison In Figure 12, we evaluate Spearman's rank correlation among different metrics and hyperparameters over 78 vision models (details in Appendix C.1). We find most metrics highly correlated with each other.

Cross-modal comparison We measure vision-language alignment using a range of alternative metrics. We visualize the corresponding alignment results in Figure 13 and Figure 14. Our findings indicate that alignment sensitivity not only depends on the metric used to compute it but also varies according to the specific tasks on which the vision models are trained.

Metric	Property				Description
	symmetric	global	ordinal	batchable	
CKA	✓	✓	✓	✓	Centered Kernel Alignment (CKA; Kornblith et al. (2019)) measures the similarity of neural networks by comparing the alignment of their kernel induced by their feature spaces.
Unbiased CKA	✓	✓	✓	✓	Unbiased estimator of CKA that corrects for sample bias in HSIC (Song et al., 2012).
SVCCA	✓	✓	✓	✓	Singular Value Canonical Correlation Analysis (SVCCA; Raghu et al. (2017)) compares neural networks by decomposing their activities into singular vectors and measuring correlation.
Mutual k-NN	✓			✓	Measures the intersection over union (IoU) of nearest neighbors between two models.
CKNNA	✓	✓$*$	✓	✓	Modified CKA measure that computes the kernel alignment only for its nearest neighbors. See Appendix A.
Cycle k-NN				✓	Measures whether the nearest neighbor in one domain also considers the original sample as its nearest neighbor in the other domain.
Edit k-NN	✓	✓$*$	✓		Computes the edit distance required to match the nearest neighbors between two datasets. The score is normalized by the maximum edit distance.
LCS k-NN	✓	✓$*$	✓		Calculates the longest common subsequence of nearest neighbors and is normalized by the sequence length.

Figure 11. Comparative analysis of neural network similarity metrics. ✓$*$ indicates the metric is global and still meaningful when the nearest neighbor k is set to maximum batch-size $k = |\mathcal{X}|$.

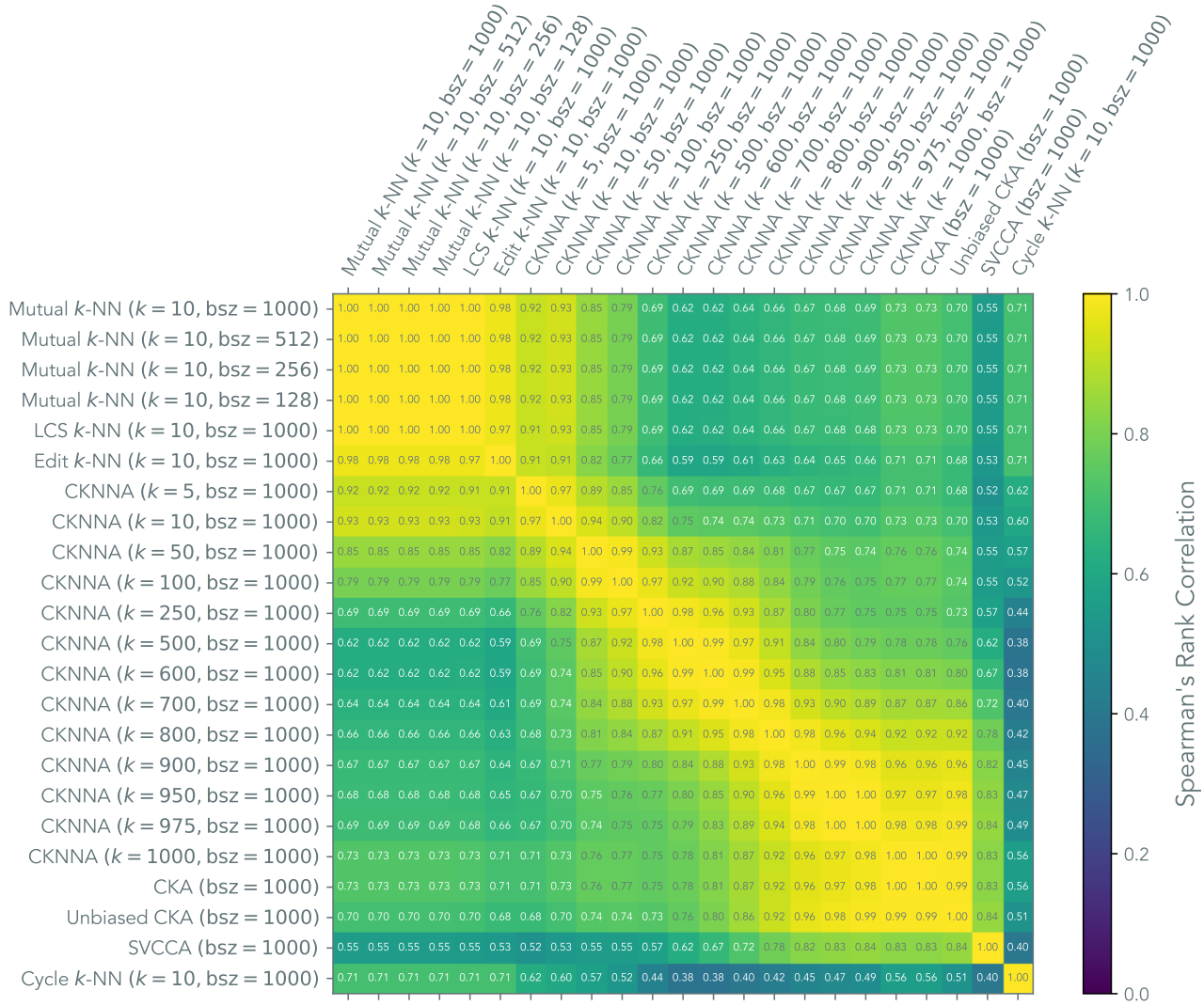

Figure 12. **Vision-vision alignment measured with various metrics.** Spearman's rank correlation among different metrics and batch sizes (bsz) when used to measure alignment among 78 vision models (see Appendix C.1 for details of these models). All p-values are below 2.24×10^{-105}. Our vision-vision analysis in Figure 2 is based on the first metric (Mutual k-NN with $k = 10$ and bsz $= 1000$).